

Qatar-1b

R-2759

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130826 · created 2026-09-03T00:14:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456530.88805 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.054
Transit depth (Rp/Rs)^2	1.2e-05 +/- 0.0038 [%]
Orbital Inclination (inc)	80.82 +/- 2.91
Airmass coefficient 1 (a1)	1.82 +/- 0.44
Airmass coefficient 2 (a2)	-0.32 +/- 0.16
Scatter in the residuals of the lightcurve fit is	26.08 %
Best Comparison Star	#3 - [86, 366]
Optimal Aperture	1.23
Optimal Annulus	6.54
Transit Duration (day)	0.024 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.054 vs 0.1463 (deviation 2.7σ)
Duration fit vs published	0.024 d vs 0.0692 d (deviation 1.73σ)
Mid-time O-C	-0.3 min
Depth SNR	0.0
Residual scatter	26.08 %

Light curve

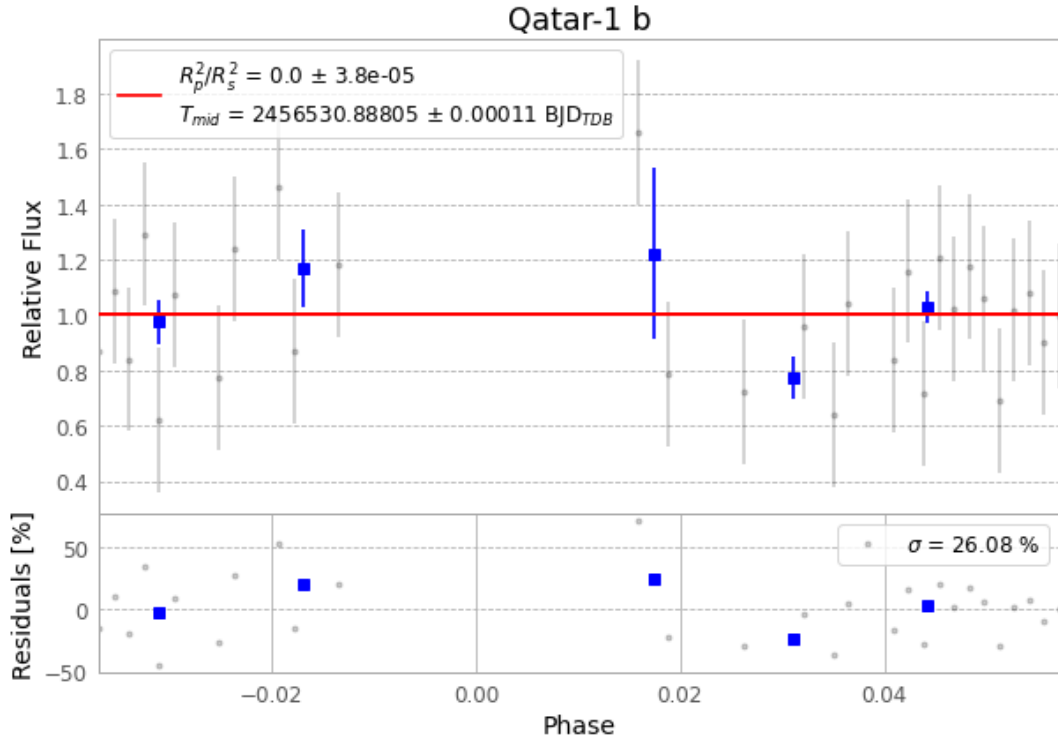


Figure 1 — FinalLightCurve_Qatar-1 b_2013-08-26.png (SHA-256 6c439a044100bcc9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [352, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 64636325f796408c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1130826080012.FITS → Qatar-1130826111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130826.log (5839f5c7d2d2...), FinalLightCurve_Qatar-1 b_2013-08-26.png (6c439a044100...), FinalLightCurve_Qatar-1 b_2013-08-26.csv (8b85e8394fac...)

Compute resources

Wall time 252.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:14Z.